6.8 PROTECTIVE DEVICES OF TRANSFORMER

The following are the protective devices in transformer and are explained below:

a) Conservator
b) Breather
c) Explosion vent
d) Buchholz relay
e) Transformer oil



Fig 6.13 *Protective Devices of Transformer*

a) Conservator

Transformer oil losses its insulating properties and is oxidised when it is in contact with the atmosphere. For this reason, the oil must not come in direct contact with the air outside. Conservators or oil expansion chambers are provided to prevent this absorption.

The conservator is cylindrical vessel. It is fitted on the top of the tank. The tank is entirely filled up with oil. The conservator is filled with oil partly (about 50%). The transformer oil gets heated due to the losses in a transformer. The volume increases due to heat and the level of oil in conservator increases. Air is expelled from the conservator through the breather. When the coil cools down, the volume decreases and the level of the oil in the conservator comes down.

This is referred to a ”breathing”. The oil surface in the conservator is only exposed to oxidation. The sludge is thus confined to the oil surface in the conservator. If there is no conservator the sludge will stick to the cooling tubes. This will spoil the cooling effort.

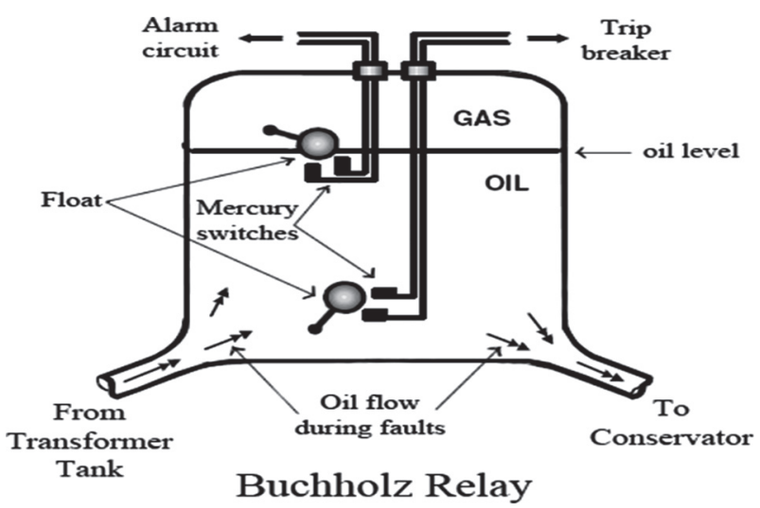


Fig 6.14 *Buchholz Relay*

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